MARKOV PROCESSES

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Contents

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Part 1. Types of Markov Processes

I follow mainly [Lam77, Chapter 6].

1. Continuous Time and Discrete Space

Assume states $\{1, \dots, N\}$.

1.1. Markov transition function $P_{ij}(t)$.

The process $X(t)$ should represent the state at time $t \geq 0$. $P(t)$ is *interpreted* as the $N \times N$ matrix such that

$$P_{ij}(t) = P(X(t) = j \mid X(t) = i) \quad t \geq 0. \tag{1}$$

Motivated by this, a *Markov transition function* is defined to be a set $(P(t))_{t \geq 0}$ of $N \times N$ matrices s.t.

$$P_{ij}(t) \geq 0 \; \forall i, j, \quad \sum\nolimits_j P_{ij}(t) = 1 \; \forall i, \quad \text{i.e. } P(t) \text{ is a } \textit{stochastic} \text{ matrix}, \tag{2}$$

$$P(0) = I_n, \tag{3}$$

$$P(t+s) = P(t)P(s) \quad \forall s, t \geq 0, \tag{4}$$

$$P(t) \text{ is right continuous at } t = 0. \tag{5}$$

1.2. Exponential.

The *exponential* of a square matrix A is

$$e^A = \exp A := \sum_{n \geq 0} \frac{A^n}{n!}. \tag{6}$$

The series converges since $A_{ij} \leq \|A\|^n$. Moreover, it is straightforward to check

$$e^{A+B} = e^A e^B \quad \text{if } AB = BA, \tag{7}$$

$$\frac{d}{dt} e^{tA} = A e^{tA} = e^{tA} A. \tag{8}$$

1.3. Generator.

Theorem 1.1. *Suppose $(P(t))_{t \geq 0}$ is a Markov transition function.*

Then $P(t)$ is differentiable, and in particular continuous, for all t. Let

$$G = \frac{d}{dt}\Big|_{t=0+} P(t) = \lim_{t \to 0+} \frac{P(t) - I}{t}. \tag{9}$$

It follows

$$P(t) = e^{tG} \quad t \geq 0, \tag{10}$$

$$G_{ij} \geq 0 \text{ for } i \neq j, \quad \sum_j G_{ij} = 0 \; \forall i. \tag{11}$$

Moreover,

$$P'(t) = P(t)\, G = G\, P(t). \tag{12}$$

Conversely, if G satisfies (11) and $P(t)$ is defined by (10) then $P(t)$ is a Markov transition function.

The correspondence between P and G is one-one.

Proof.

1: *Continuity* of $P(t)$ for all t follows from (4) and (5).

2: To see $P(t)$ is *everywhere differentiable* note

$$\big(P(h) - I\big)\big(I + P(h) + \cdots + P((n-1)h)\big) = P(nh) - I,$$

$$\therefore \frac{P(h) - I}{h}\, h\big(I + P(h) + \cdots + P((n-1)h)\big) = P(nh) - I.$$

Fix $t > 0$, let $h \to 0$ and $n \to \infty$ s.t. $nh \to t$. Then[1]

$$\lim_{h\to 0^+} \frac{P(h) - I}{h} = (P(t) - I)\left(\int_0^t P(u)\,du\right)^{-1},$$

using continuity of P and the Riemann sum approximation to the integral.

3: Define G as in (9). Then (11) is clear. Moreover $P'(t)$ exists for all t and satisfies (12) since $P'(0)$ exists and by (4).

Since $P(t)$ satisfies (12), and by standard ODE theory the unique solution which equals I at $= 0$ is e^{tG}, it follows (10) is true.

4: Finally suppose G is as in (11). Then we claim $P(t) := e^{tG}$ is a Markov transition matrix. Then (3), (4) and (5) are clear.

For (2) we first note the rows of $P(t)$ sum to 1 since

$$e^{tG}\mathbf{1} = I\mathbf{1} = \mathbf{1},$$

because the rows of G sum to 0.

It remains to prove $P(t) \geq O$, the matrix of zeros.

First assume $G_{ij} > 0$ for $i \neq j$. Then for small $t > 0$

$$P(t) = I + tG + o(t) > 0.$$

Hence for *arbitrary* $t > 0$, $P(t) = (P(t/k))^k > O$, by taking k sufficiently large.

If we only have $G \geq 0$, let E_ϵ be the matrix with all entries $\epsilon > 0$. Then

$$P = \lim_{\epsilon\to 0} e^{G+E_\epsilon} \geq O,$$

since $e^{G+E_\epsilon} > O$ by the previous argument. □

1.4. **Random Clock.** The natural probabilistic way to generate a continuous time Markov chain $(P(t)_{t\geq 0}$ is via a discrete time Markov chain Q governed by a random clock following a Poisson R.V. with parameter λ.

More precisely, transitions occur at times given by a Poisson R.V. with mean λt in time intervals of length t. Each transition is governed by a transition matrix Q.

Thus the probability of k transitions in $[0, t]$ is $e^{-\lambda t}(\lambda t)^k/k!$, and the probability of passing from state i at time 0 to state j by time t is

$$P_{ij}(t) = e^{-\lambda t}\sum_{k\geq 0} \frac{(\lambda t)^k}{k!} Q^k_{ij}. \tag{13}$$

Equivalently,

$$P(t) = e^{\lambda t(Q-I)}. \tag{14}$$

In particular, *the generator of* $(P(t))_{t\geq 0}$ *is* $\lambda(Q - I)$.

Conversely, if $P(t)$ is a given transition function with generator G then we can describe the process in the above manner by taking

$$Q = I + \lambda^{-1}G, \quad \lambda \geq \max_i |G_{ii}|. \tag{15}$$

The reason for the restriction on λ is so that $0 \leq Q_{ii} \leq 1$. Recall $-1 \leq G_{ii} \leq 0$ from (11).

2. Discrete Time and General State Space

2.1. **Markov kernel** $P(x, E)$**.** Consider a measurable state-space $(S, \mathcal{S})$.

$P(x, E)$ is *interpreted* as *the probability of moving from* x *into* E *in one transition.*

Motivated by this, a *Markov kernel* or *stochastic kernel* is defined to be a function $P(x, E)$ where $x \in S$ and $E \in \mathcal{S}$, such that $P(x, E)$ is a probability measure on $(S, \mathcal{S})$ for fixed x, and is measurable in x for fixed E.

[1]From the theorem, the integral on the right side is

$$\int_0^t e^{uG}\,du = G^{-1}(e^{tG} - I) = G^{-1}(P(t) - I).$$

2.2. **Properties.** Define

$$
\begin{aligned}
P^0(x,E) &= \delta_x, \\
P^1(x,E) &= P(x,E), \\
P^{n+1}(x,E) &= \int P(x,dy)P^n(y,E).
\end{aligned} \tag{16}
$$

Then P^n gives the n-step probabilities.

It follows readily that

$$
P^{n+m}(x,E) = \int P^n(x,dy)\,P^m(y,E). \tag{17}
$$

2.3. **Redistributions.** Instead of probabilities or probability distributions $P(x,\cdot)$, it is often helpful to think of a mass distribution ν_x and its redistributions.

Define

$$
\begin{aligned}
&\nu_x(E) = P(x,E),\ \text{ i.e. } \nu_x \text{ is the redistribution of } \delta_x \text{ after one step,} \\
&\nu_{n,x}(E) = P^n(x,E),\ \nu_{n,x} \text{ is the redistribution of } \delta_x \text{ after } n \text{ steps,} \\
&\nu_\mu(E) = \int P(x,E)\,d\mu(x) = \int \nu_x(E)\,d\mu(x),\ \nu_\mu \text{ is the redistribution of } \mu \text{ after one step.}
\end{aligned} \tag{18}
$$

Note

$$
P(x,dy) = d\nu_x(y), \quad P^n(x,dy) = d\nu_{n,x}(y).
$$

Then (17) becomes

$$
\nu_{m+n,x}(E) = \int d\nu_{n,x}(y)\,\nu_{m,y}(E).
$$

That is, the redistribution of δ_x after $n+m$ steps equals the redistribution after n steps *of* the redistribution of δ_x after m steps.

3. Continuous Time and State Space

3.1. **Markov transition function $P_t(x,E)$.** Consider a measurable state-space $(S,\mathcal{S})$.

$P_t(x,E)$ is *interpreted as the probability that a system in state x at time s will move to some state in E at time $s+t$.*

Motivated by this, a *Markov transition function* is defined to be a function $P_t(x,E)$ where $t \geq 0$, $x \in S$ and $E \in \mathcal{S}$. The requirements are that $P_t(x,E)$ be a probability measure on $(S,\mathcal{S})$ for fixed t and x, and measurable in x for fixed t and E. Moreover, we require

$$
\begin{aligned}
P_0(x,E) &= \mathcal{X}_E(x) \quad (\text{i.e. } P_0(x,\cdot) = \delta_x), \\
P_{t+s}(x,E) &= \int_S P_t(x,dy)\,P_s(y,E) \quad \text{if } s,t \geq 0.
\end{aligned} \tag{19}
$$

The last equation is the *Chapman Kolmogorov* equation and reflects the Markov principle of lack of "memory".

In terms of *mass distributions* as in Section 2.3, set

$$
\nu_{t,x}(E) = P_t(x,E).
$$

Then the Chapman Kolmogorov equation becomes

$$
\nu_{t+s,x}(E) = \int d\nu_{t,x}(y)\,\nu_{s,y}(E),
$$

and says that the redistribution of δ_x at time $t+s$ equals the redistribution at time t of the redistribution of δ_x at time s.

3.2. **Brownian motion.** Suppose a probability measure μ is *infinitely divisible*.[2] It follows that for each $t > 0$ there is a probability measure μ_t such that $\mu_1 = \mu$ and $\mu_{s+t} = \mu_s * \mu_t$.

It follows that for $S = \mathbb{R}$ and $\mathcal{S} = \mathcal{B}$ (Borel sets on $\mathbb{R}$),

$$P_t(x, E) := \mu_t(E - x)$$

defines a translation invariant measure. That is $P_t(x, E) = P_t(x + y, E + y)$. The proof is essentially notational.[3]

It follows that the process has *stationary independent increments*.[4]

The most important example is $\mu = N(0, 1)$, the normal distribution with mean 0 and variance 1.[5] Then $\mu_t = N(0, t)$ (variance t), i.e.

$$P_t(x, E) = \frac{1}{\sqrt{2\pi t}} \int_{E-x} e^{-y^2/2t} dy. \tag{20}$$

This is the transition function for *Brownian motion* [BM].

The *continuity condition*

$$P_t(x, [x - \epsilon, x + \epsilon]) = 1 - o(t) \quad \epsilon > 0 \tag{21}$$

is clearly satisfied. This implies the process moves by continuous motion. Such a markov process is called a *diffusion*.[6]

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[2]That is, for each integer $n > 0$ there is a probability measure μ_n for which

$$\mu = \mu_n * \cdots * \mu_n \ n \text{ times}.$$

Recall $\mu * \nu = \nu * \mu$ and

$$\mu * \nu(E) = \iint \mathcal{X}_E(x + y)\, d\mu(x)\, d\nu(y) = \int \mu(E - y)\, d\nu(y).$$

Recall also that if $\operatorname{dist} X = \mu$, $\operatorname{dist} Y = \nu$, and X and Y are independent, then $\operatorname{dist}(X + Y) = \mu * \nu$.

[3]One has for $t, s > 0$,

$$\begin{aligned} P_{t+s} &= \mu_{t+s}(E - x) = \mu_s * \mu_t(E - x) \\ &= \int \mu_s(E - x - y)\, d\mu_t(y) = \int d\mu_t(y - x)\, \mu_s(E - y) \end{aligned}$$

($y \mapsto y - x$, and where $\mu_t(y - x)$ is translation of $d\mu_t$ to right by x)

$$= \int P_t(x, dy)\, P_s(y, E).$$

For this to be true at $t = 0$ it is clearly necessary and sufficient that $P_0(x, \cdot) = \delta_x$.

[4][Lam77, p125] says "independent increments". But this is true for any Markov process.

[5]Recall that if $\operatorname{dist} X = N(0, s)$ and $\operatorname{dist} X = N(0, t)$, and X and Y are independent, then $\operatorname{dist}(X + Y) = N(0, s + t)$.

[6]If one defines a markov process by using a stochastic kernel $Q(x, E)$ and a random clock analogous to the manner in Section 1.4 then one instead obtains a *jump* process.

Part 2. Semigroups

I follow mainly [Lam77, Chapter 7].

4. Construction of Semigroups from Markov Processes

4.1. **Notation.** $P_t(x,E)$ is a Markov transition function on $(S,\mathcal{S})$ as in Section 3. In particular note (19).

Define the Banach spaces

$$F(S) = \{f : S \to \mathbb{R} \mid \text{ is bdd and measurable}\} \text{ with the sup norm } \|f\|, \tag{22}$$

$$C(S) = \{f : S \to \mathbb{R} \mid \text{ is bdd and continuous}\} \text{ with the sup norm } \|f\|, \tag{23}$$

$$M(S) = \{\mu \mid \mu \text{ is a finite signed measure on } S\} \text{ with the variation norm } \|\mu\|, \tag{24}$$

where $\|\mu\| = \mu^+(S) + \mu^-(S)$.

4.2. **Contraction semigroups on $F(S)$ and $M(S)$.** (This is a 'warm up". The important case is Section 4.3.)

Define

$$T_t : F(S) \to F(S), \quad T_t f(x) = \int f(y) P_t(x,dy), \tag{25}$$

$$U_t : M(S) \to M(S), \quad U_t\mu(E) = \int d\mu(x)\, P_t(x,E) = \int d\mu(x)\, \nu_{t,x}(E). \tag{26}$$

Thus if X_t is the process corresponding to $P(x,E)$ then we have the interpretation

$$T_t f(x) = \mathbb{E}^x f(X_t),$$

Analogous to the third equation in (18), we interpret $U_t\mu$ as the redistribution $\nu_{t,\mu}$ of μ at time t.

Since $U_t\mu$ is a measure it operates on $f \in F(S)$, and using dual pairing notation we have

$$\begin{aligned}
\langle U_t\mu, f\rangle &:= U_t\mu(f) \\
&= \int d\mu(x) \left(\int P_t(x,dy)\, f(y) \right) \text{ since from (26) we have the case } f = \mathcal{X}_E \\
&= \int d\mu(x)\, T_t f(x) = \langle \mu, T_t f\rangle.
\end{aligned}$$

That is,

$$\langle U_t\mu, f\rangle = \langle \mu, T_t f\rangle. \tag{27}$$

This can be interpreted for μ a probability measure as follows: The mean of f w.r.t. the redistributed μ equals the mean obtained by choosing x via μ, starting the process from x and evaluating f at X_t, i.e. evaluating $\mathbb{E}\big(\mathbb{E}^x f(X_t)\big)$.

It is easy to check[7] that T_t and U_t define contraction semigroups, i.e.

$$\|T_t f\| \le \|f\|, \quad T_{t+s} = T_t \circ T_s, \tag{28}$$

$$\|U_t\mu\| \le \|\mu\|, \quad U_{t+s} = U_t \circ U_s. \tag{29}$$

In order to have a useful theory, we will need the *continuity condition* which, in the case of $F(S)$, is

$$\lim_{t\to 0^+} \|T_t f - f\| = 0 \quad \forall f \in F(S).$$

[7] In particular,

$$\begin{aligned}
T_{t+s}f(x) &= \int P_{t+s}(x,dy)\, f(y) \text{ by definition} \\
&= \iint P_t(x,du)\, P_s(u,dy) f(y) \text{ by the Chapman Kolmogorov eqn (19)} \\
&= \int P_t(x,du) T_s f(u) = (T_t \circ T_s f)(x) \text{ by definition.}
\end{aligned}$$

To show U_t is contractive split μ into μ^+ and μ^-. Inequality may occur as some of the redistribution of μ^+ and μ^- may cancel.

This is equivalent to[8]

$$\lim_{t\to 0^+} P_t(x,\{x\}) = 1 \quad \forall x \in S.$$

However, this is not true for B.M.

4.3. Contraction semigroups on $C(S)$. The Markov transition function $P_t(x,E)$ is *Feller* if

$$T_t : C(S) \to C(S), \tag{30}$$

where $C(S)$ is the space of bounded continuous functions with the sup norm.

This is the same as requiring weak convergence of measures:[9]

$$P_t(x,\cdot) \rightharpoonup P_t(x_o,\cdot) \text{ as } x \to x_0, \quad \forall t \geq 0. \tag{31}$$

Note there is no continuity requirement here on t, which we now consider.

Henceforth we assume S is compact.

Processes on $\mathbb{R}$ and $\mathbb{R}^n$ will be studied by considering the one point compactifications $\mathbb{R}^*$ and $\mathbb{R}^{n*}$.

The transition function $P_t(x,E)$ is *uniformly stochastically continuous* if

$$\lim_{t\to 0^+} P_t(x,B_\epsilon(x)) = 1 \quad \forall \epsilon > 0, \text{ uniformly for } x \in S. \tag{32}$$

It is straightforward to check[10] that this implies

$$\lim_{t\to 0^+} \|T_t f - f\| = 0, \quad \forall f \in C(S). \tag{33}$$

The transition function $P_t(x,E)$ is *normal*[11] if it is Feller and uniformly stochastically continuous.

Remark: For a Feller transition function on a compact state space, stochastic continuity (i.e. (32) without the uniform requirement in x) implies uniform stochastic continuity and hence normality.[12]

BM on $\mathbb{R}^n$ is normal (acting on the compactified space).

5. General Contraction Semigroups

5.1. Definitions. Assume $(V, \|\cdot\|)$ is a real Banach space.

Examples to keep in mind for the following are (22) and (24).

A family $(T_t)_{t\geq 0}$ of (bounded) linear operators $T_t : V \to V$ is a *contractive semigroup* if

$$\begin{aligned} &\|T_t x\| \leq \|x\| \ \forall x \in V, \\ &T_0 = I, \ T_{t+s} = T_t \circ T_s \ \forall t,s \geq 0, \\ &\lim_{t\to 0^+} \|T_t x - x\| = 0 \ \forall x \in V. \end{aligned} \tag{34}$$

The last condition is equivalent to continuity at 0 in the *strong* topology. This is weaker than continuity in the *uniform* topology, which normally does not hold. But in this context strong continuity is equivalent to continuity in the *weak* topology.

It also follows that T_t is *uniformly strongly continuous*[13] for $t \in [0,\infty)$. The proof is straightforward from the semigroup property.[14]

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[8]Straightforward, see [Lam77, pp 154,5].

[9]Since

$$(T_t f)(x) = \int P_t(x,dy)\, f(y)\, dy = P_t(x,\cdot)(f).$$

[10]See [Lam77, PP 157,8]. The point is that since S is compact, f is uniformly continuous.

[11]Non standard notation from [Lam77].

[12]See [Lam77, pp 159–161].

[13]That is

$$\forall x\ \forall \epsilon > 0\ \exists \delta = \delta(x,\epsilon) \text{ s.t. } t \in [0,\infty),\ |s| < \delta \Longrightarrow \|T_{t+s}x - T_t x\| < \delta.$$

[14]See [Lam77, pp 137,138.]

5.2. **Infinitesimal generator.** The infinitesimal generator A of $(T_t)_{t\geq 0}$ is defined by

$$Ax = \lim_{t\to 0^+} \frac{Tx - x}{t}. \tag{35}$$

The *domain* $\mathcal{D}_A$ of A is the set of $x \in V$ for which the limit exists.

See Section 1.3, where A is the matrix G.

Proposition 5.1. *For any contractive semigroup $(T_t)_{t\geq 0}$, $\mathcal{D}_A$ is dense in V.*

Proof. Suppose $x \in V$.

The idea is to first show that if $f(s)$ is C^1 with compact support in $(0,\infty)$, then

$$y(x) := \int f(s) T_s x \, ds \in \mathcal{D}_A$$

Then take a sequence of such $f_n(s) \geq 0$ with unit integral converging to δ_0.

The first claim follows from

$$\begin{aligned} Ay &= \lim_{t\to 0^+} \frac{T_t y - y}{t} \\ &= \lim_{t\to 0^+} \int f(s) \frac{T_{t+s}x - T_s x}{t} ds \\ &= \lim_{t\to 0^+} \int \frac{f(s-t) - f(s)}{t} T_s x \, ds \\ &= \int f'(s) T_s x \, ds \quad \text{since } f \text{ is } C^1. \end{aligned}$$

Hence $y \in \mathcal{D}_A$.

Now take f_n with supports in $(0, 1/n)$ converging to $\{0\}$, $\int f_n = 1$, $f_n \geq 0$. Then

$$\|y_n(x) - x\| = \left\| \int f_n(s)(T_s x - x)\, ds \right\| \leq \int f_n(s) \|T_s x - x\| \, ds \to 0.$$

Here we used the fact $T_s x \to x$ as $s \to 0$. □

We now make the crucial observation that $T_t x$ satisfies a first order linear ODE.

In the following the limit giving the derivative is in the norm topology. This is always the case unless o.w. stated.

Proposition 5.2. *If $x \in \mathcal{D}_A$ then $T_t x \in \mathcal{D}_A$ for all $t > 0$. Moreover,*

$$\frac{d}{dt} T_t x = A(T_t x) = T_t(Ax). \tag{36}$$

The proof is standard using the semigroup property.[15]

5.3. **Bounded infinitesimal generator.** If A is bounded (this is *not* usually the case) then $x \in \mathcal{D}_A$ for *every* x , and by standard methods

$$e^{tA} x := \left(\sum_{n\geq 0} \frac{t^n A^n}{n!} \right) x$$

is well defined. Moreover, this is the *unique* solution[16] of

$$\frac{dy(t)}{dt} = Ay(t), \quad y(0) = x.$$

[15] See [Lam77, pp 139,140].

[16] Suppose $y(t)$ is any solution. Then we compute

$$\frac{d}{dt}\left(e^{-tA} y(t)\right) = 0, \quad e^{-0A} y(0) = x.$$

It follows that $e^{-tA} y(t) = x$ for all $t \geq 0$.

The last "follows" is because $u(t) \in V$ continuous on $[0, a)$ with zero derivative on $(0, a)$ implies $u(t) = u(0)$ on $[0, a]$. To see this consider arbitrary elements v of V^*, differentiate $\langle u(t) - u(0), v\rangle$ and apply the standard real valued result to deduce that $\langle u(t) - u(0), v\rangle = 0$ for all $v \in V^*$, and hence $u(t) = u(0)$.

It then follows from (36) that

$$T_t x = e^{tA} x \quad \forall t \geq 0 \text{ and } \forall x, \qquad T_t = e^{tA} \quad \forall t \geq 0.$$

5.4. **Laplace transform analogy.** If the infinitesimal generator A is unbounded then $e^{tA}x$ need not converge, even for $x \in \mathcal{D}_A$.

So we try to solve the linear ODE (36) by modelling the approach on the Laplace transform method.

Recall that for bounded real valued functions $x : \mathbb{R}^+ \to \mathbb{R}$ the Laplace transform is

$$(\mathcal{L}x)(\lambda) = X(\lambda) := \int_0^\infty e^{-\lambda t} x(t)\, dt, \quad \lambda \geq 0.$$

Then differentiation and integration become multiplication and division by λ.

Integration by parts gives

$$(\mathcal{L}x')(\lambda) = \lambda X(\lambda) - x(0).$$

The inverse transform we need is

$$X(\lambda) = \frac{1}{\lambda - a}\ (\lambda > a) \Longleftrightarrow x(t) = e^{-at}\ (t \geq 0).$$

So if we want to solve the following ODE (the analogue of (36))

$$\frac{dx(t)}{dt} = ax(t), \quad x(0) = x_0,$$

then translating this into an algebraic equation for the Laplace transform we get

$$\begin{aligned} \lambda X(\lambda) - x_0 &= aX(\lambda) \\ \therefore\ X(\lambda) &= (\lambda - a)^{-1} x_0 \quad \lambda > 0 \\ \therefore\ x(t) &= e^{-at} x_0 \quad t \geq 0. \end{aligned}$$

The above will motivate the approach to solving (36) by using resolvents.

References

[Lam77] John Lamperti, *Stochastic processes*, Springer-Verlag, 1977. A survey of the mathematical theory; Applied Mathematical Sciences, Vol. 23.